Knowledge Graphs

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Lab Introduction
Who are we?

Motivation
Why learn this?

Organization
How to get my mark?

Content
Representations

Content
Systems

Content
Applications

Knowledge Graphs

The aim of this is course is to gain a deep understanding of Knowledge Graphs divided into three blocks. An overarching aim of the course is to understand the connections between Knowledge Graphs, Artificial Intelligence, Machine Learning and Data Science.

Representations

Logical and ML-based representations for KGs.

Read more details

KG Embeddings
Widely-applied, large family of ML models.

Logical Knowledge in KGs
Highly expressive, diverse family of logical models.

Graph Neural Networks
Using the KG structure *as* a neural network.

Data Models
Overview of data models in different communities.

Systems

Systems to bring KGs into practice.

Read more details

Architectures
The big picture of building IT architectures for KGs.

Scalable Reasoning
Making use of the knowledge in the KG.

KG Creation
How to create a KG from heterogeneous data?

KG Evolution
How to update, correct and complete a KG?

Applications

Real-world applications of KGs.

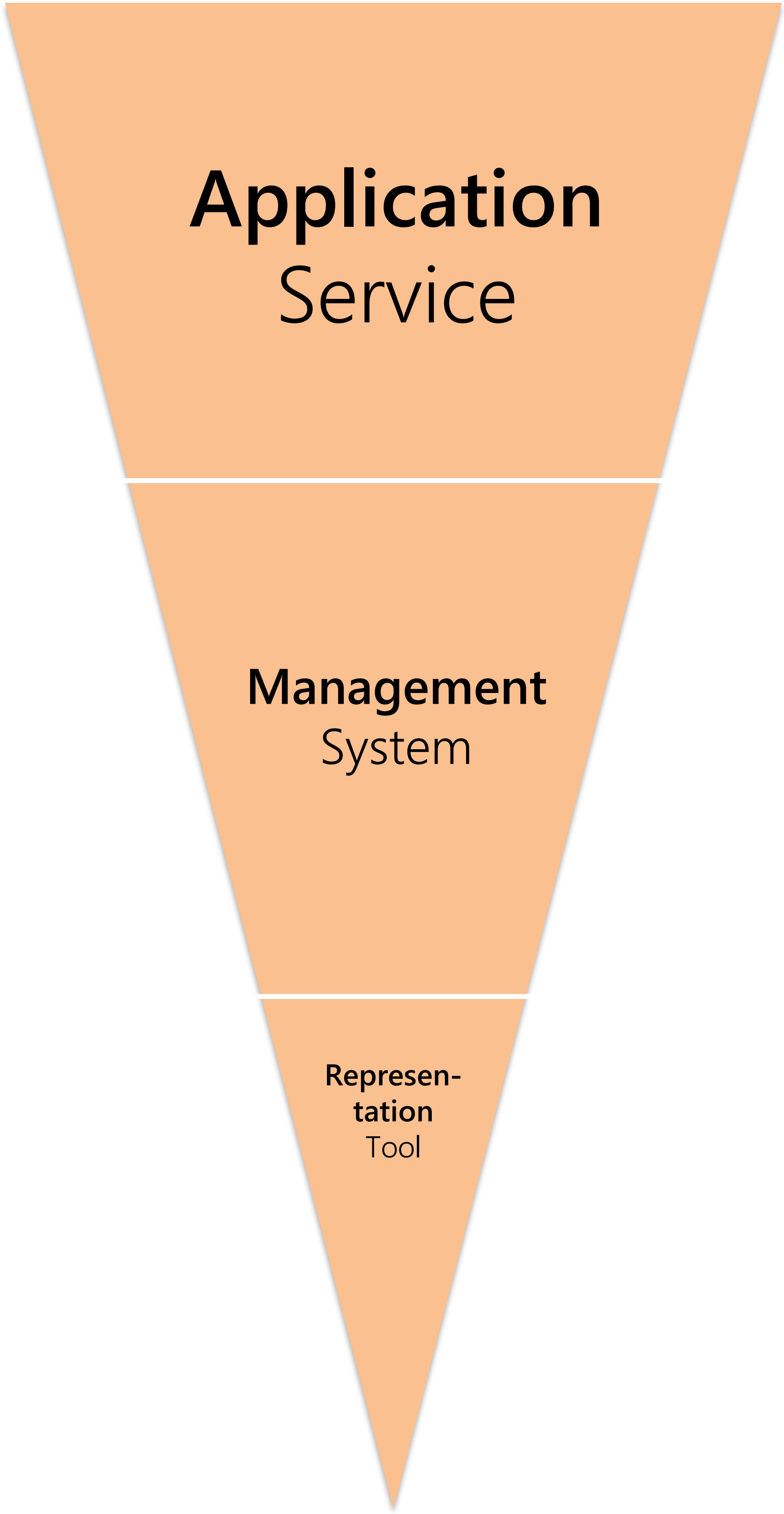
Read more details

Real-World Applications
Overview of diverse applications.

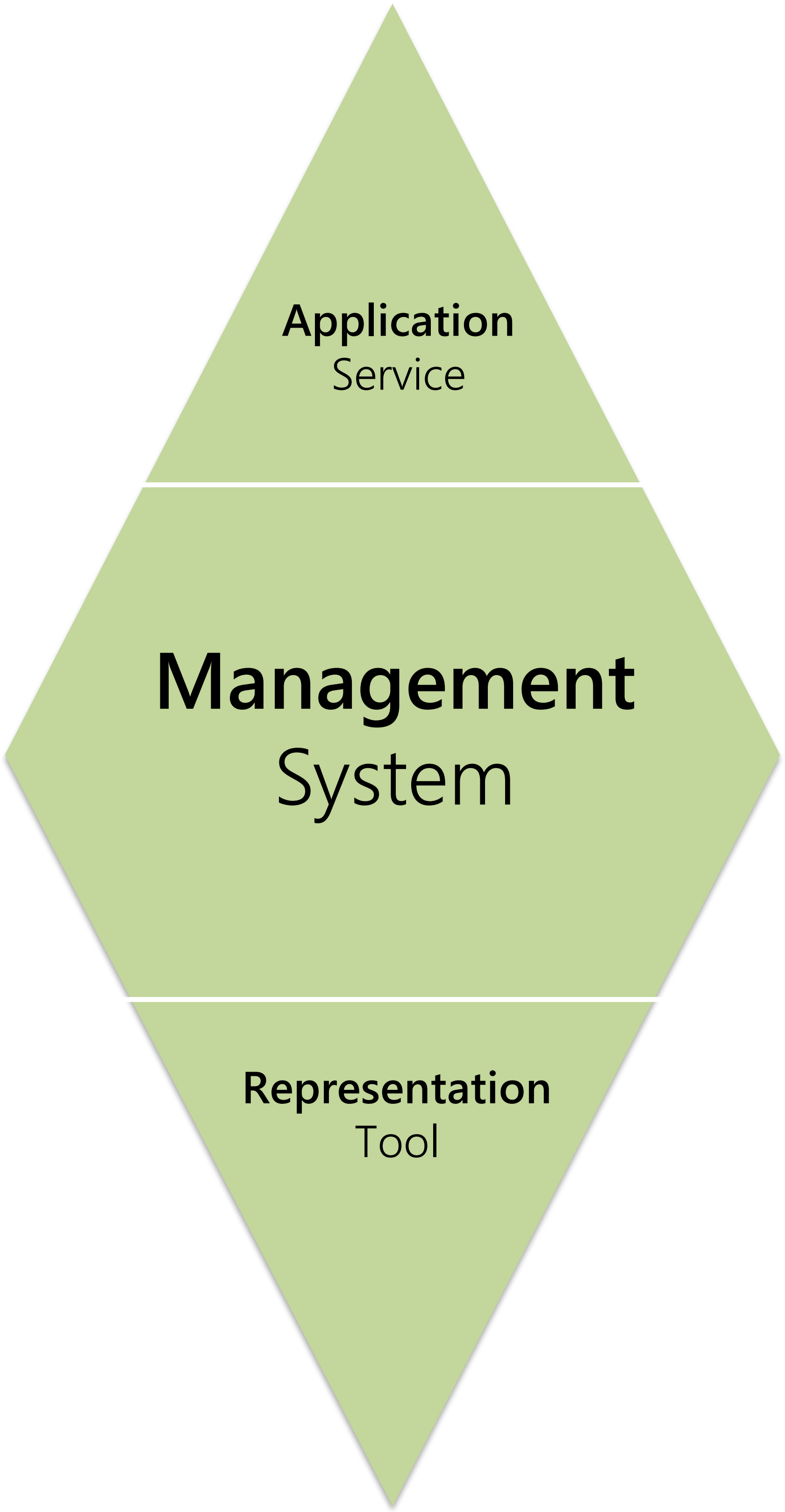
Financial KGs
Concrete applications in finance and economics.

Services
Which service to provide based on KGs?

Connections
... between KGs, AI, ML and Data Science.



Application
Service
Management
System
Represen-
tation
Tool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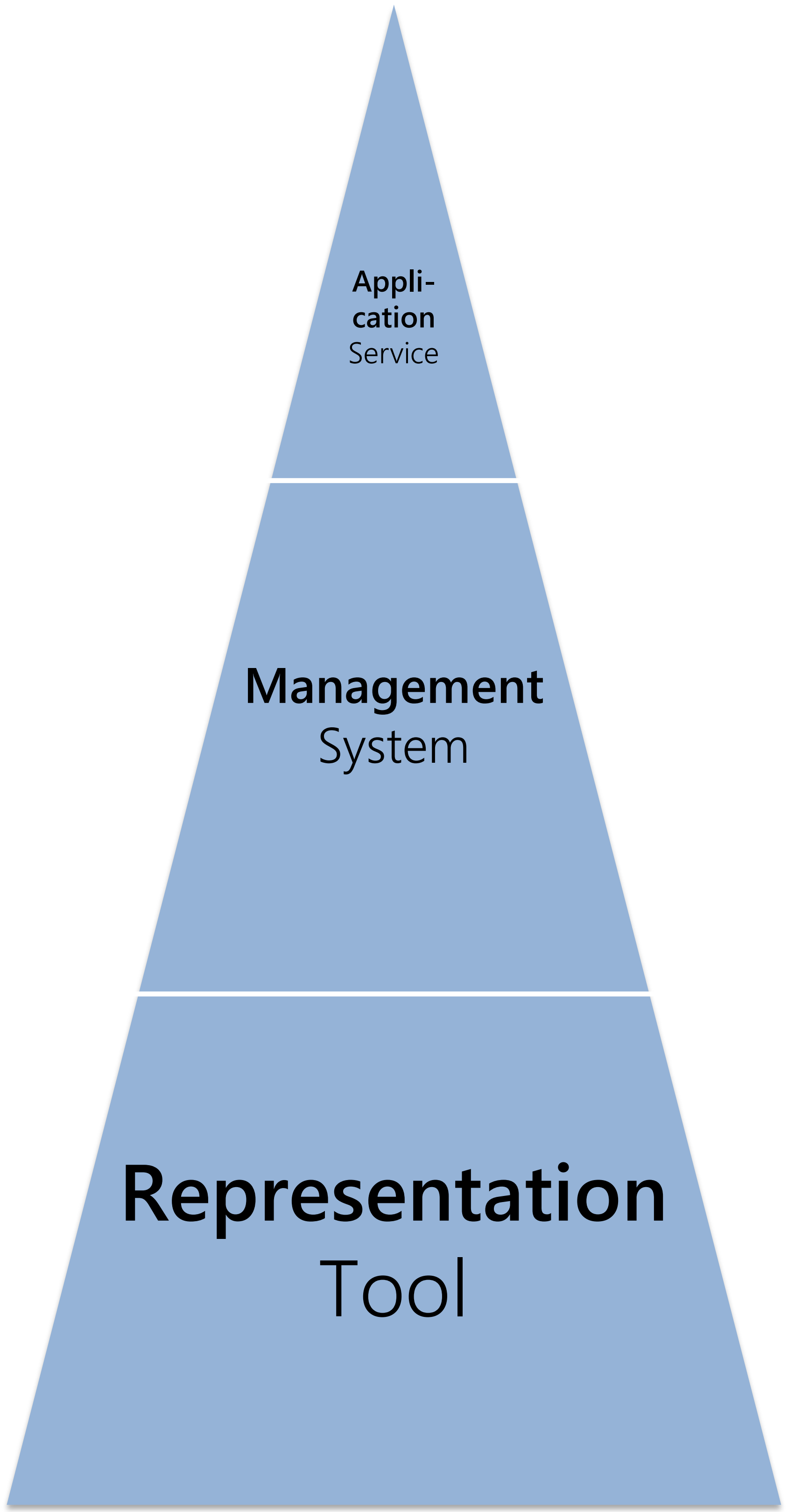
Appli-
cation
Service
Management
System
Representation
Tool

What is a Knowledge Graph?
Surveying the Space

What is a Knowledge Graph?
Choosing a Definition

"A **knowledge graph** acquires and integrates information into an **ontology** and applies a **reasoner** to derive new knowledge"

"Towards a Definition of Knowledge Graphs". L. Ehrlinger and W. Wöß. SEMANTICS 2016.

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Knowledge Graph

$$KG = (D, K; g, l, r)$$

Obtaining Learning

Applying Reasoning

Relational Data

...

Property Graphs

XML

JSON

RDF*

CSV Files

RDF

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